Project Id:-39
"OncoScan AI "

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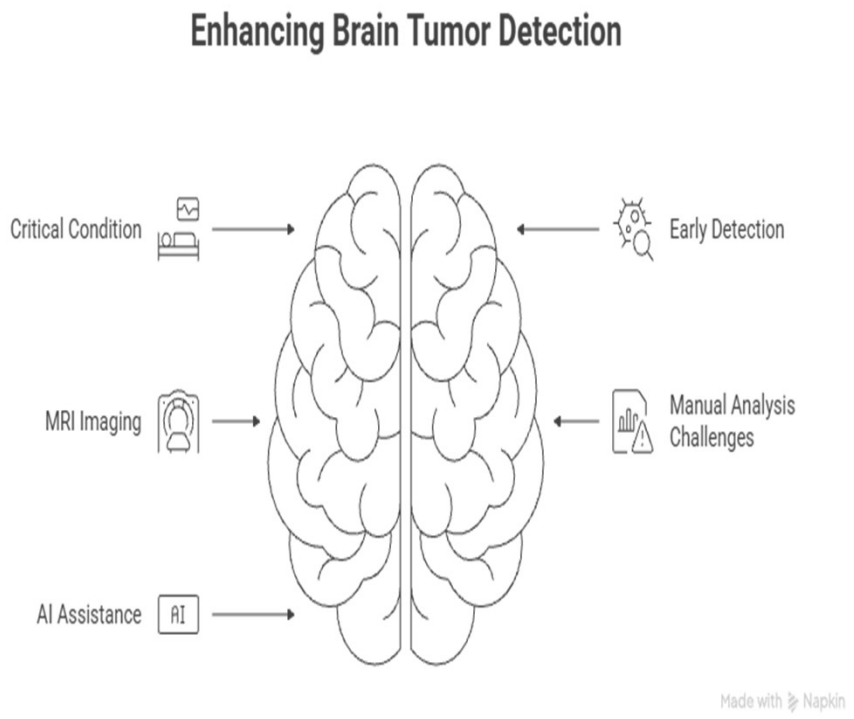
Index

- **Introduction** 1
- **Objectives** 2
- **Literature Survey** 3
- **Problem Statement** 4
- **Proposed Solution** 5
- **Conclusion** 6
- **References** 7

Introduction

Brain tumors are one of the most critical and life-threatening medical conditions affecting the human brain. Early and accurate detection is essential for effective treatment and improving patient survival rates. Magnetic Resonance Imaging (MRI) is widely used for diagnosing brain tumors due to its high-resolution imaging capabilities. However, manual analysis of MRI scans by radiologists is time-consuming, subjective, and prone to human error.

The proposed system aims to assist healthcare professionals by providing a fast, reliable, and automated solution for brain tumor detection, ultimately supporting early diagnosis and better clinical decision-making.



Objectives

The main objectives of the proposed **OncoScan AI** system are:

•To develop an automated system for **brain tumor detection** using MRI images.

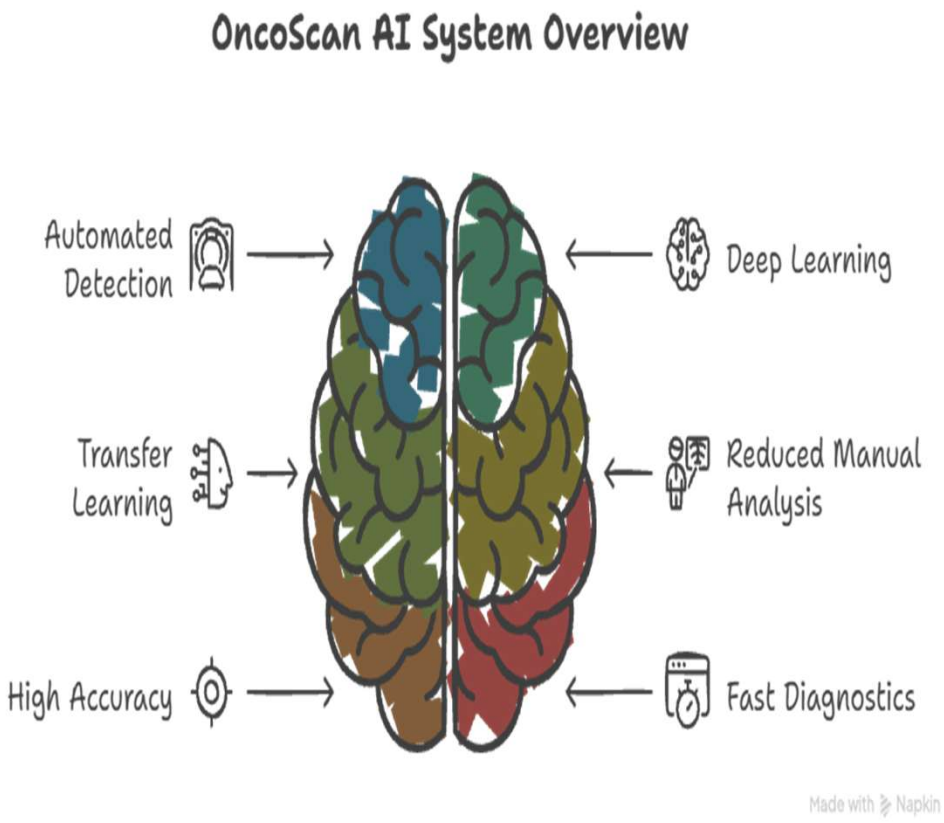
•To utilize **deep learning techniques**, specifically the Deep Learning model **ResNet50**, for accurate classification.

•To apply **transfer learning** for improving performance on limited medical datasets.

•To reduce the dependency on **manual analysis by radiologists** and minimize human error.

•To achieve **high accuracy, precision, and reliability** in tumor detection.

•To provide a **fast and efficient diagnostic support tool** for healthcare professionals.

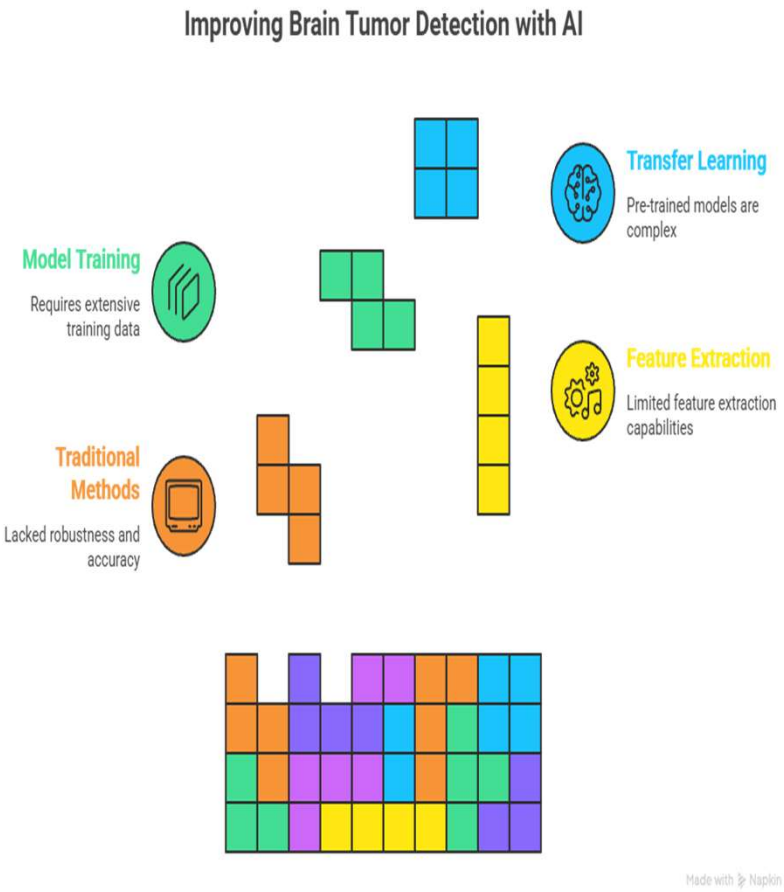


Litrature Survey

Recent advancements in **Artificial Intelligence** and **Deep Learning** have significantly improved the accuracy of brain tumor detection using MRI images. Various research studies have explored different approaches for automated medical image analysis.

Early methods were based on traditional image processing and basic machine learning techniques, which achieved moderate accuracy but lacked robustness. With the introduction of **Convolutional Neural Networks (CNNs)**, the performance improved significantly due to better feature extraction capabilities.

Several studies have used pre-trained models such as **VGG16**, **ResNet50**, and Inception networks for tumor classification. These models, when combined with **transfer learning**, achieved high accuracy ranging from **95% to 99%**.

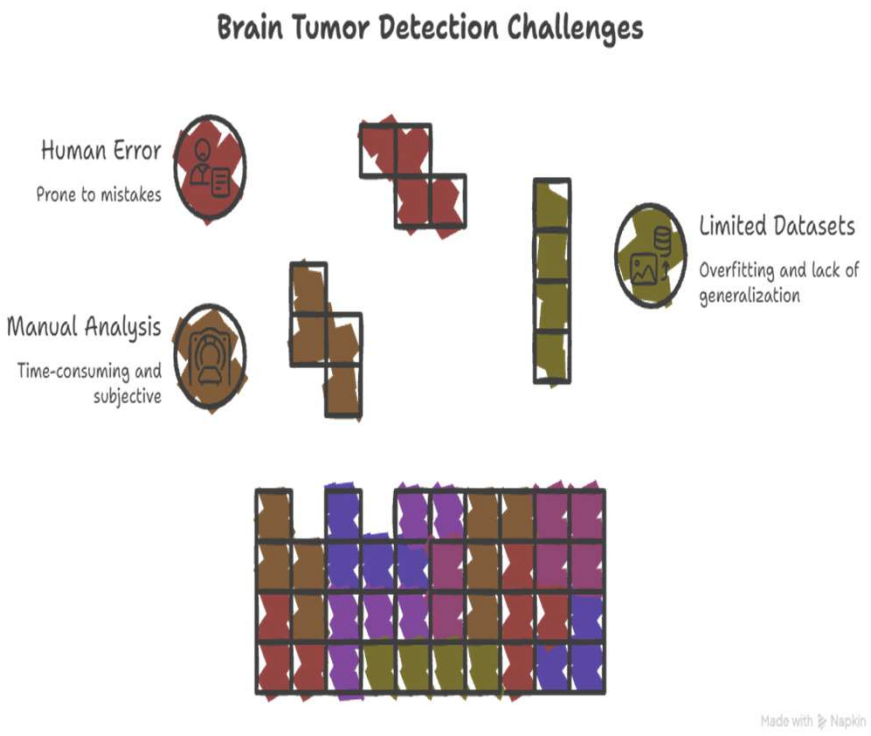


Problem Statement

- Brain tumor detection using MRI images is a complex and critical task in medical diagnosis. Traditional methods rely heavily on manual analysis by radiologists, which is time-consuming, subjective, and prone to human error.

- Existing automated systems often face challenges such as limited dataset availability, overfitting, and lack of generalization across different MRI scans.

- Therefore, there is a need to develop an intelligent and efficient system that can automatically analyze MRI images and accurately detect the presence of brain tumors with minimal human intervention.

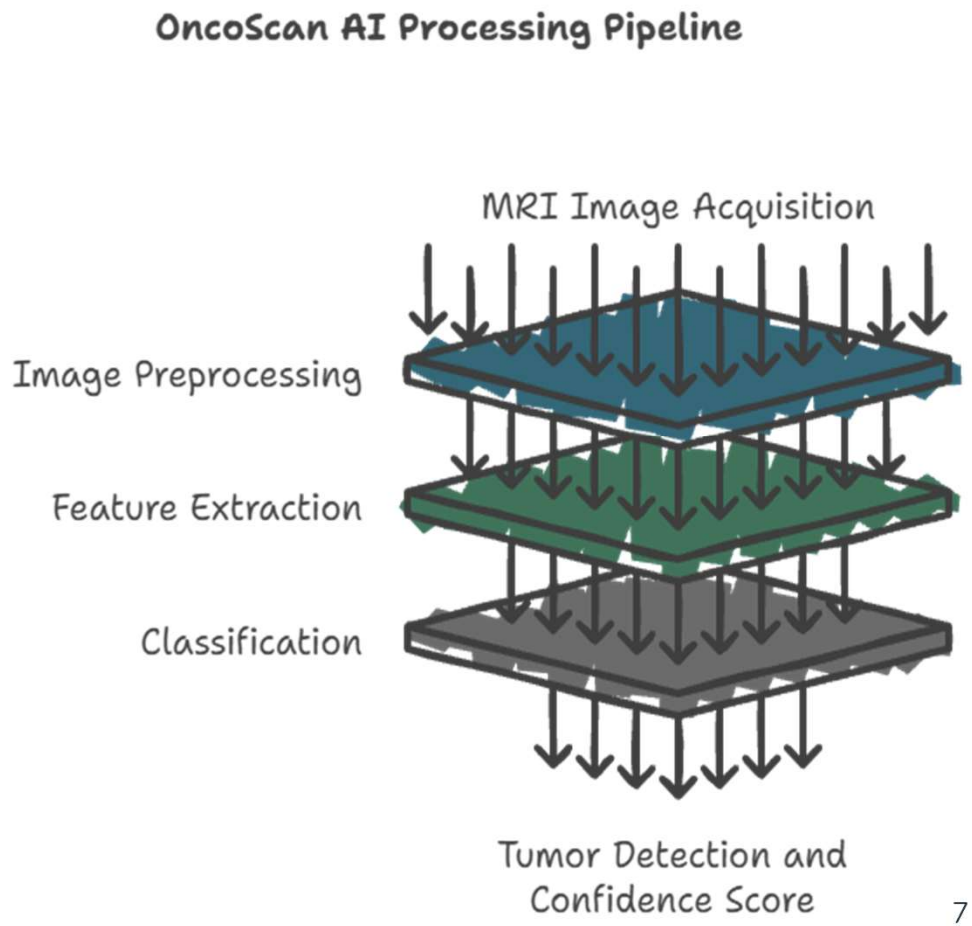


Proposed Solution

- The proposed system, **OncoScan AI**, is an automated brain tumor detection model that utilizes advanced **Deep Learning** techniques to analyze MRI images efficiently and accurately.

- The system follows a structured pipeline starting from image acquisition to final prediction. MRI images are first preprocessed using techniques such as resizing, normalization, and data augmentation to enhance image quality and improve model performance.

- A pre-trained **ResNet50** model is used as the core feature extractor. Through **transfer learning**, the model leverages previously learned features and is fine-tuned on brain MRI datasets to improve accuracy while reducing training time.



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Conclusion

- In this project, **OncoScan AI** demonstrates the effective use of **Deep Learning** techniques for automated brain tumor detection using MRI images. By leveraging the power of the **ResNet50** model along with transfer learning, the system achieves high accuracy and reliability even with limited datasets.

- The proposed solution reduces the dependency on manual analysis and helps overcome challenges such as time consumption and human error in traditional diagnosis methods. It provides a fast, efficient, and scalable approach that can assist healthcare professionals in early detection and decision-making.

- Although the system shows promising results, it is designed as a supportive tool and not a replacement for medical experts. With further improvements such as multi-class classification, real-time integration, and explainable AI, the system can be enhanced for practical deployment in healthcare environments.

References

•Research on brain tumor detection using **Deep Learning** techniques and MRI images.

•Studies on **transfer learning** applications in medical image analysis.
•ResNet50 architecture for image classification tasks.
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•Brain tumor classification using Convolutional Neural Networks (CNNs).

•Medical image processing techniques using **TensorFlow** and Keras.

•Open-source MRI datasets and research papers available on platforms like IEEE Xplore and Google Scholar.

Thankyou!